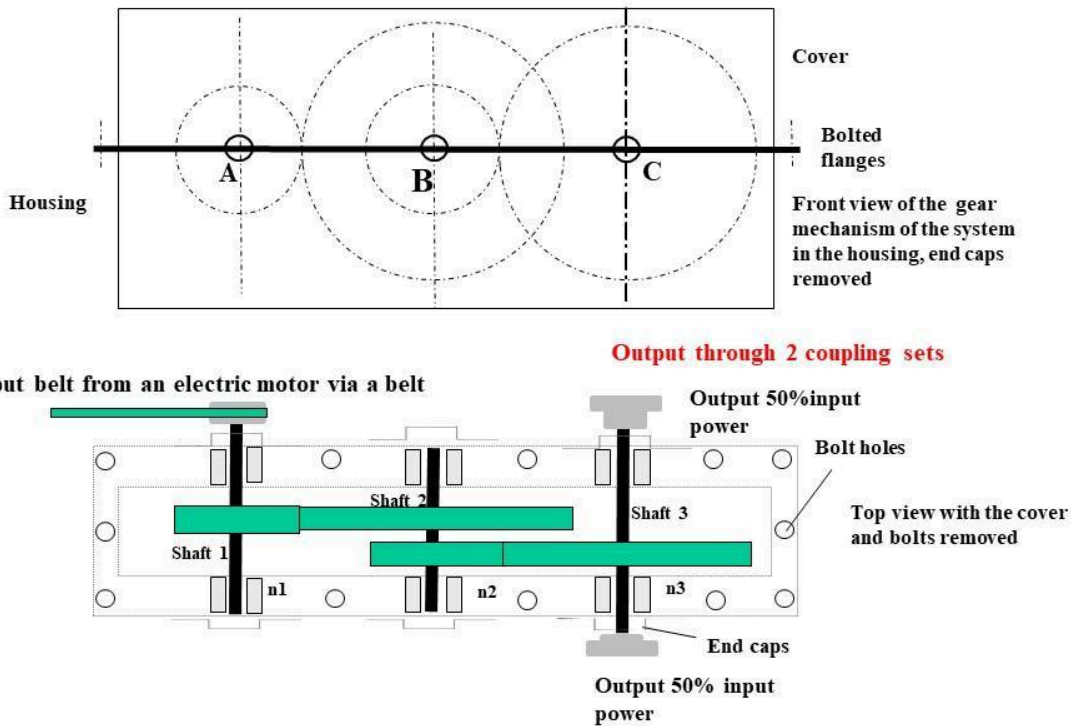


ME 315
THEORY OF MACHINES – DESIGN OF ELEMENTS
 Fall 2025

Design Project
 Group 8 Submission



Group members

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Shaft system 1
 Shaft system 2
 Shaft system 3

Grading

On-time step dues	(5)
Structural design, drawings	/40 (assembly, parts, dimensions, tolerances, etc.)
Design analysis	/40 (shaft, gear(s), bearings)
Demonstration/Report	/15 (view quality, cover sheet, page numbers, clarity, logic, conciseness)

Total _____/95

Basic Gear Box Diagram

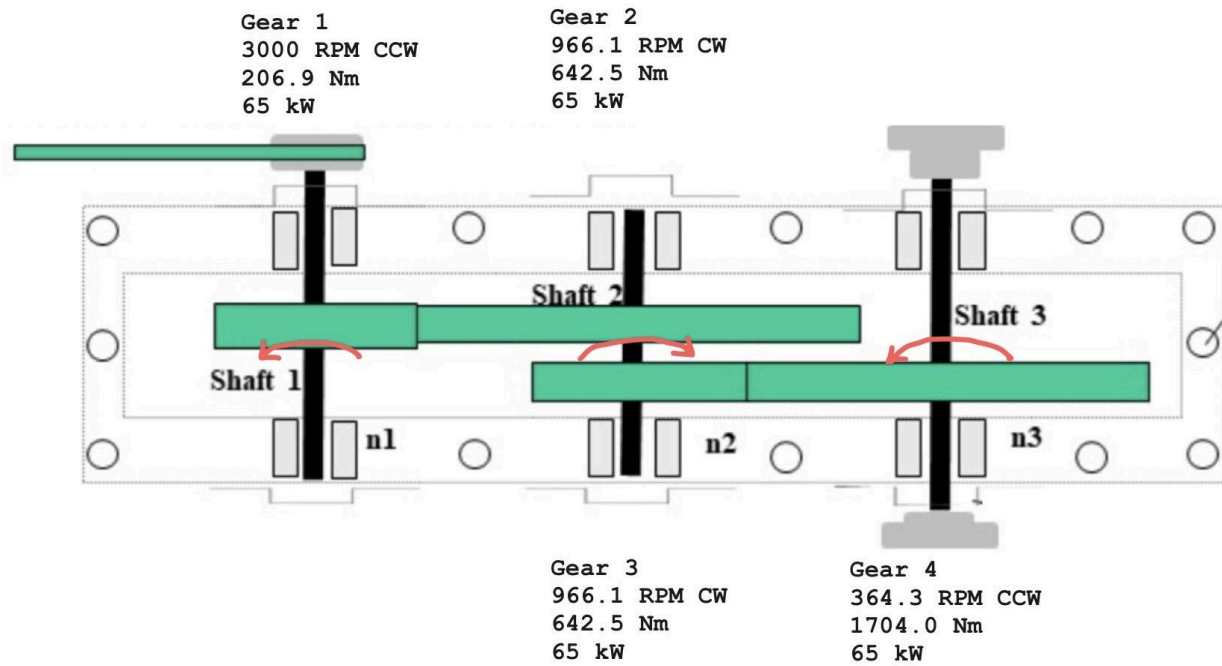


Table 1: Geartrain and Gear Parameters

Item	Gear 1	Gear 2	Gear 3	Gear 4
Teeth (N)	19	59	23	61
Module (mm)	5	5	5	5
Pitch Diameter (mm)	95	295	115	305
Pressure Angle (°)	20°	20°	20°	20°
Face Width (mm)	60	60	60	60
Type	Spur	Spur	Spur	Spur
Speed (rpm)	3000.0	966.1	966.1	364.3
Total Torque (Nm)	206.9	642.5	642.5	1704.0
Power (kW)	65	65	65	65
Tangential Force at Pitch Circle (N)	4356	4356	11174	11174
Radial Force (N)	1585	1585	4067	4067
Material	4140 Grade 2	4140 Grade 1	4140 Grade 2	4140 Grade 2
Hardness (HB)	450	375	450	375
Hardness Ratio	1.2		1.2	
Bending FoS	6.08	3.39	1.94	1.7
Contact FoS	3.15	1.20	1.12	1.12
Contact Ratio	1.576		1.518	
Stage Transmission Ratio	0.32203		0.37705	
Overall Transmission Ratio	0.12142			
Center Distance (mm)	195		210	
Total Center Distance (mm)	405			
Transmission Error	0			

Table 2: Shafts and Parameters

Item	Shaft 1	Shaft 2	Shaft 3
Material	1045	8620	8620
Speed (rpm)	3000.0	966.1	364.3
Max Torque T (Nm)	206.9	642.5	1704.0
Max Bending Moment M (Nm)	236	516.3	454.2
# of Cycles (MC)	2621	844.0	318.2
Location of Critical Cross Section	Center of Gear 1	Center of Gear 3	Center of Gear 4
Static FoS	6.43	6.80	5.3
Fatigue FoS	1.92	1.63	1.56
Expected Life	Infinite	Infinite	Infinite

Table 3: Bearings and Parameters

Item	Bearing A & B	Bearing C & D	Bearing E & F
Max Load (N)	5070	9387	8258
Calculated C (N)	69903	70862	56379
SKF Bearing #	6410	NU 210 ECM	6310
Inner Diameter (mm)	50	50	50
Outer Diameter (mm)	130	90	110
Width (mm)	31	20	27
C (N)	87100	73500	61800
Co (N)	52000	69500	38000
Expected Life (Millions of Cycles)	2621	953.3	318.25
Bearing Change Services	Not Needed	Not Needed	Not Needed

Table 4: Overall Results

Item	Result
Power (kW)	65
Input Speed (rpm)	3000
Output Speed (rpm)	364
Lowest Factor of Safety	1.12
Maintenance Needs	Lubrication and oil changes periodically

Shaft Assembly Drawings

Shaft 1 Assembly

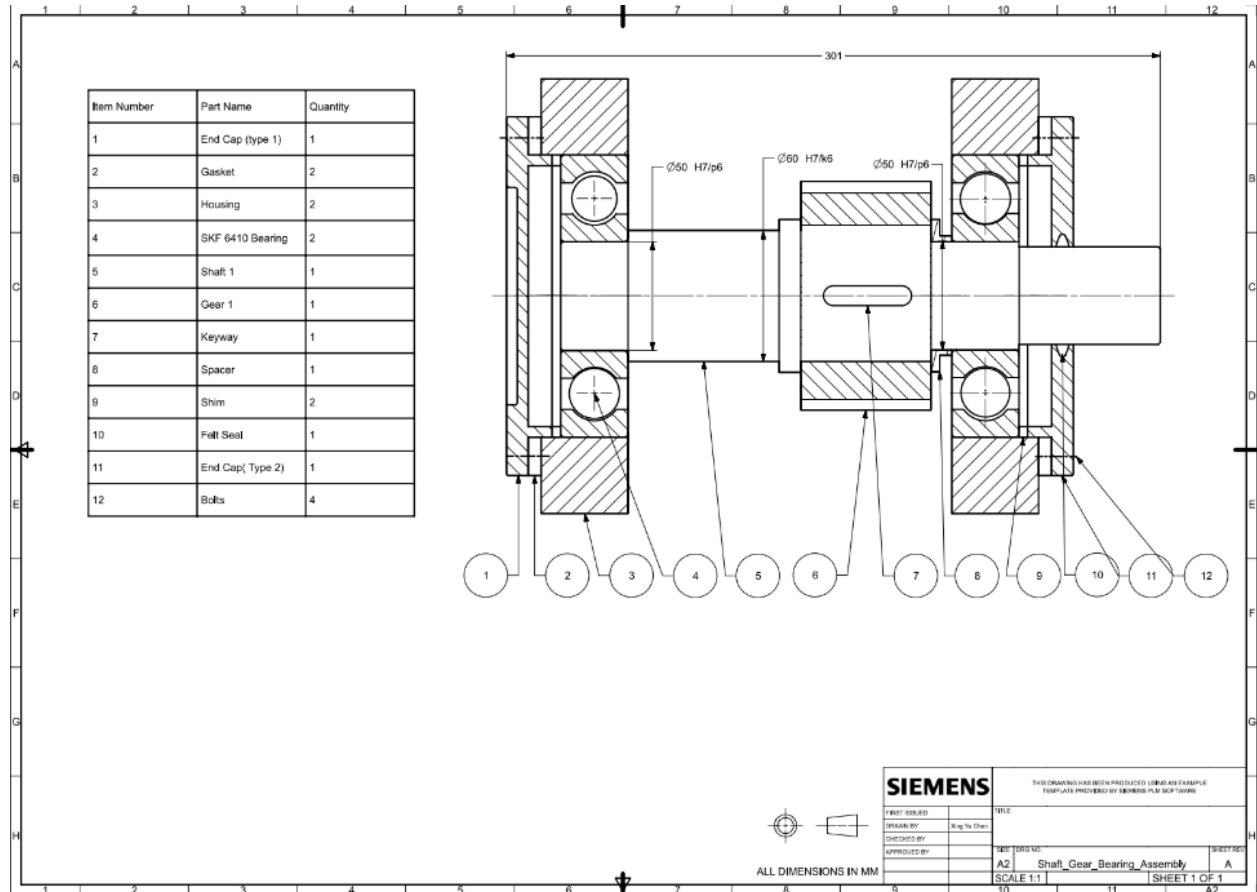
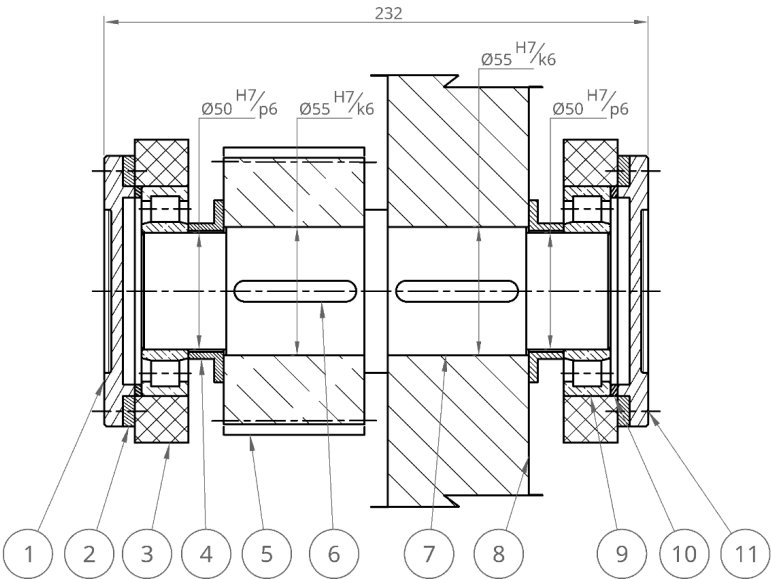


Figure 1: Shaft 1 Assembly

Shaft 2 Assembly

ITEM NO.	PART NAME	QTY.
1	End Cap	2
2	Gasket	2
3	Housing	2
4	Spacer	2
5	Gear 3	1
6	Shaft Keys	2
7	Shaft 2	1
8	Gear 2	1
9	SKF NU 210 ECM Roller Bearing	2
10	Shim	2
11	Bolts	4



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS ANGULAR - ° '	DRAWN	NAME	SIGNATURE	DATE	TITLE		
	CHECKED	MASON MESSMER		2025-12-04			
	APPROVED				Shaft 2 Assembly		
	DO NOT SCALE DRAWING						
BREAK ALL SHARP EDGES AND REMOVE BURRS					Shaft 2		
FIRST ANG. & PROJECTION	MATERIAL		FINISH				
					SIZE	DWG NO.	REV.
					A4		
					SCALE	1:2	1 of 1

Figure 2: Shaft 2 Assembly

Shaft 3 Assembly

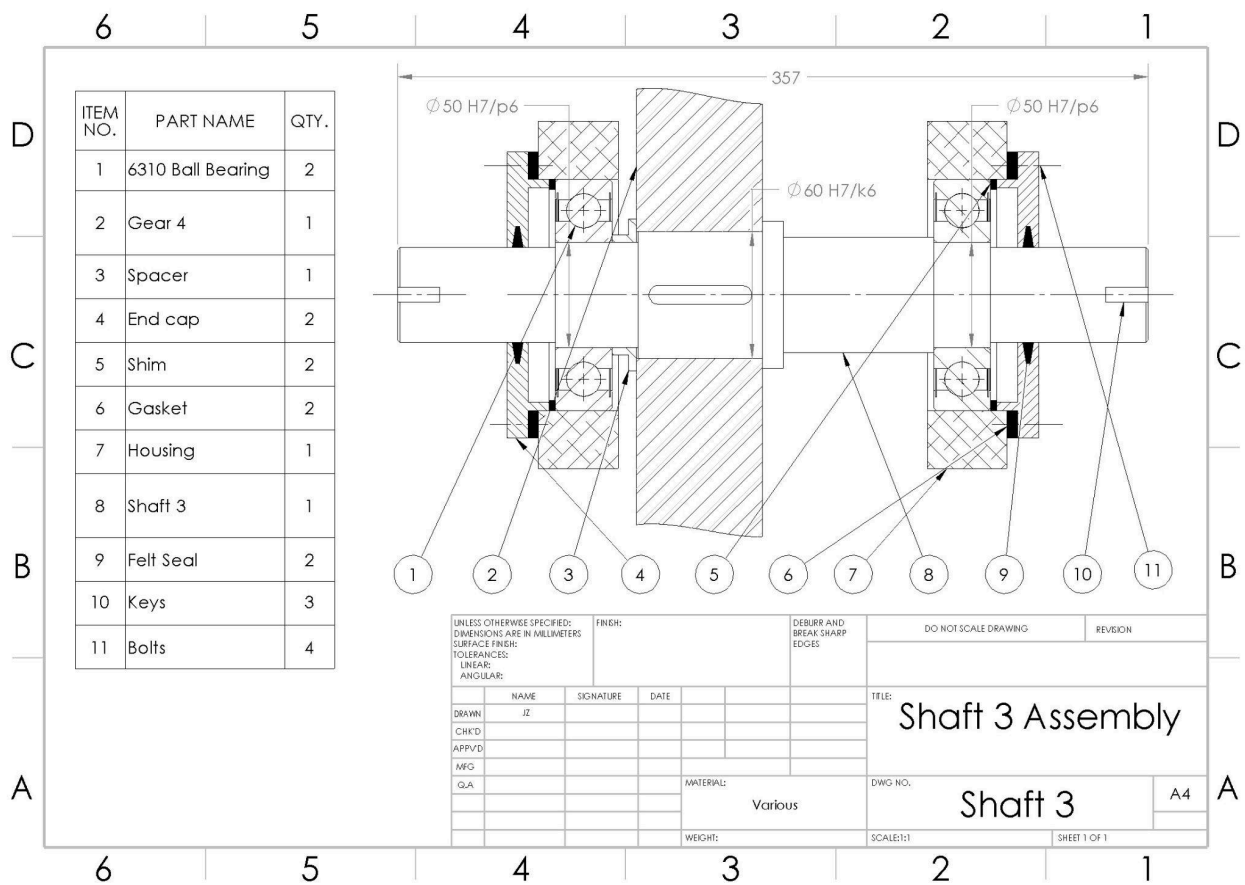


Figure 3: Shaft 3 Assembly

Shaft Part Drawings

Shaft 1

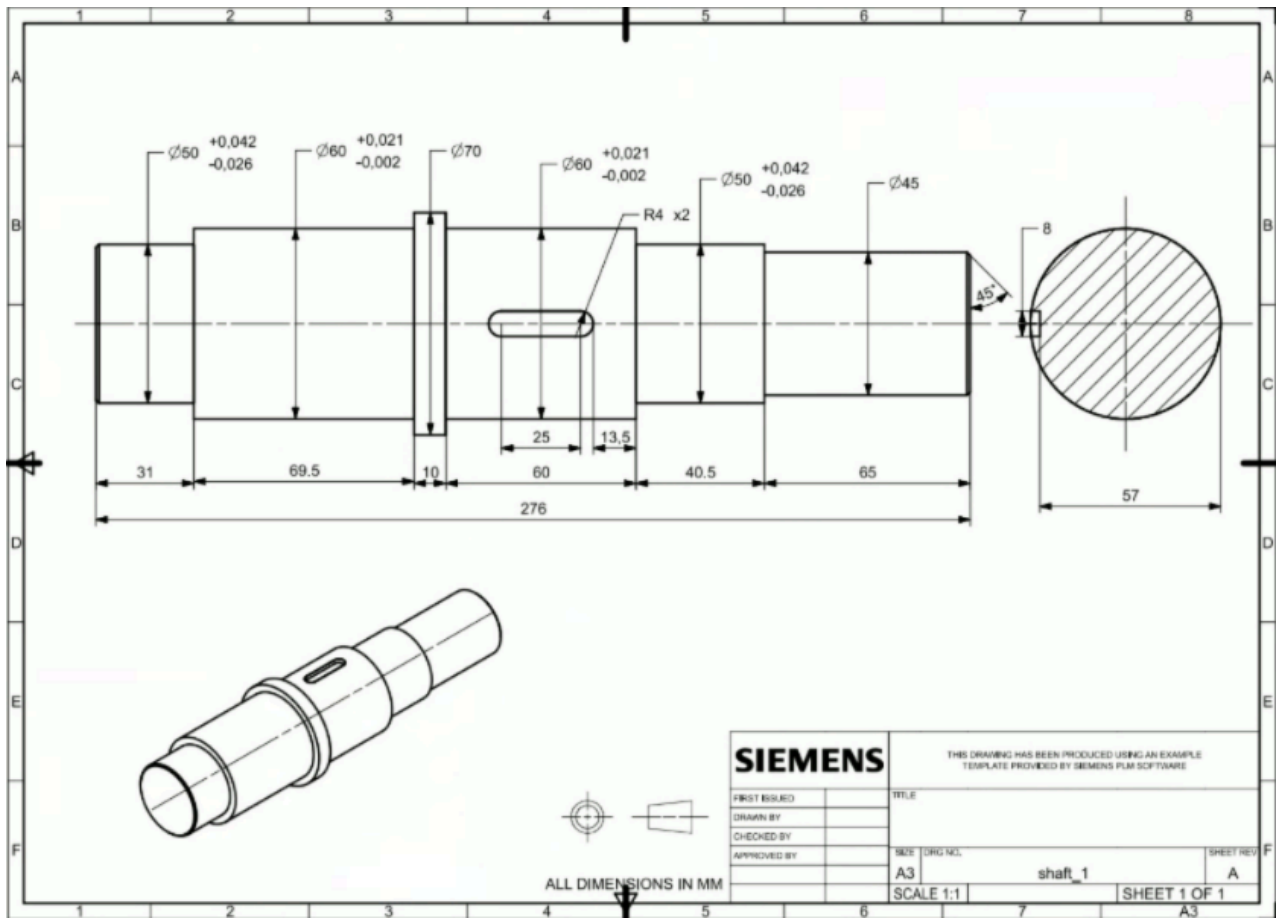


Figure 4: Shaft 1

Shaft 2

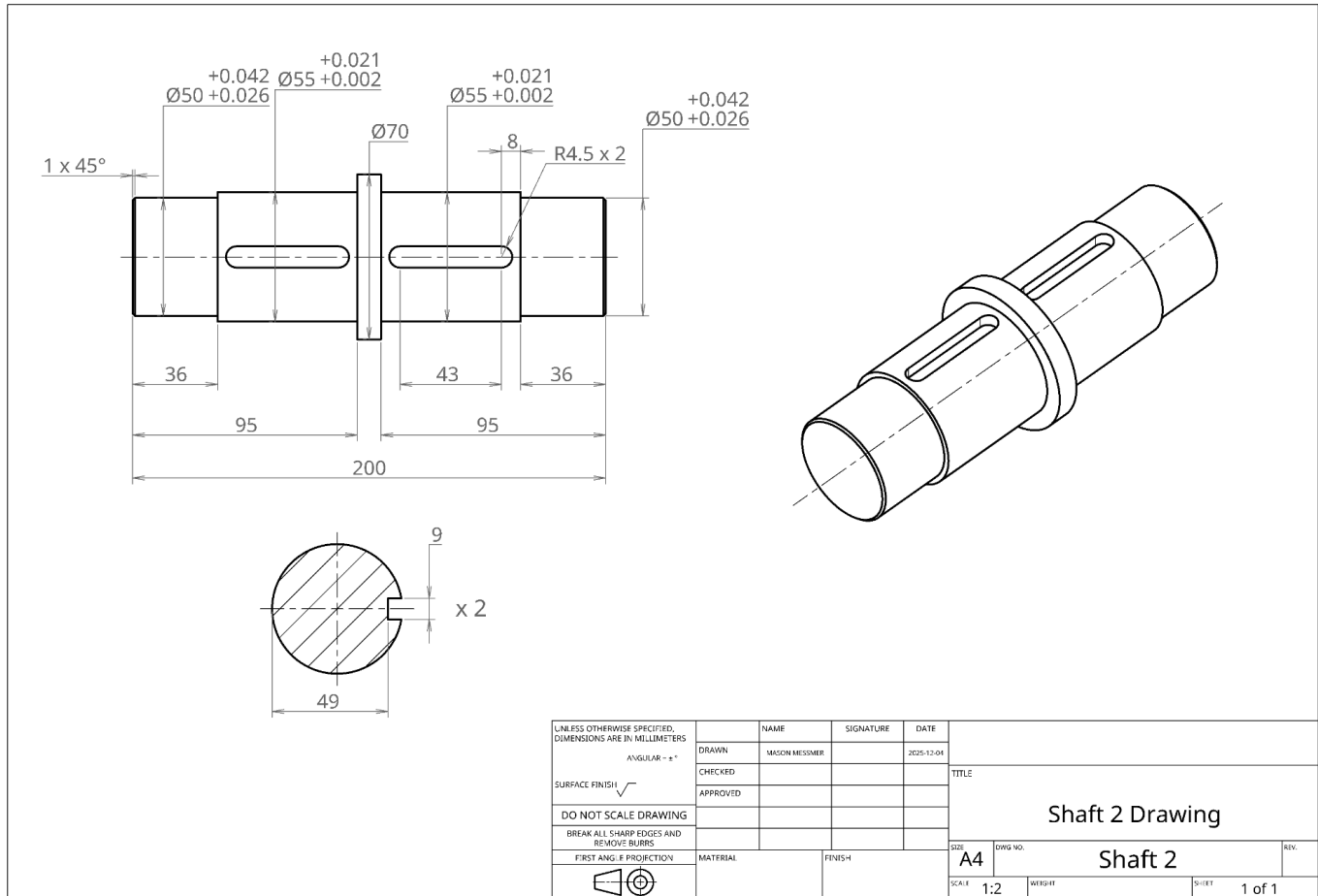


Figure 5: Shaft 2

Shaft 3

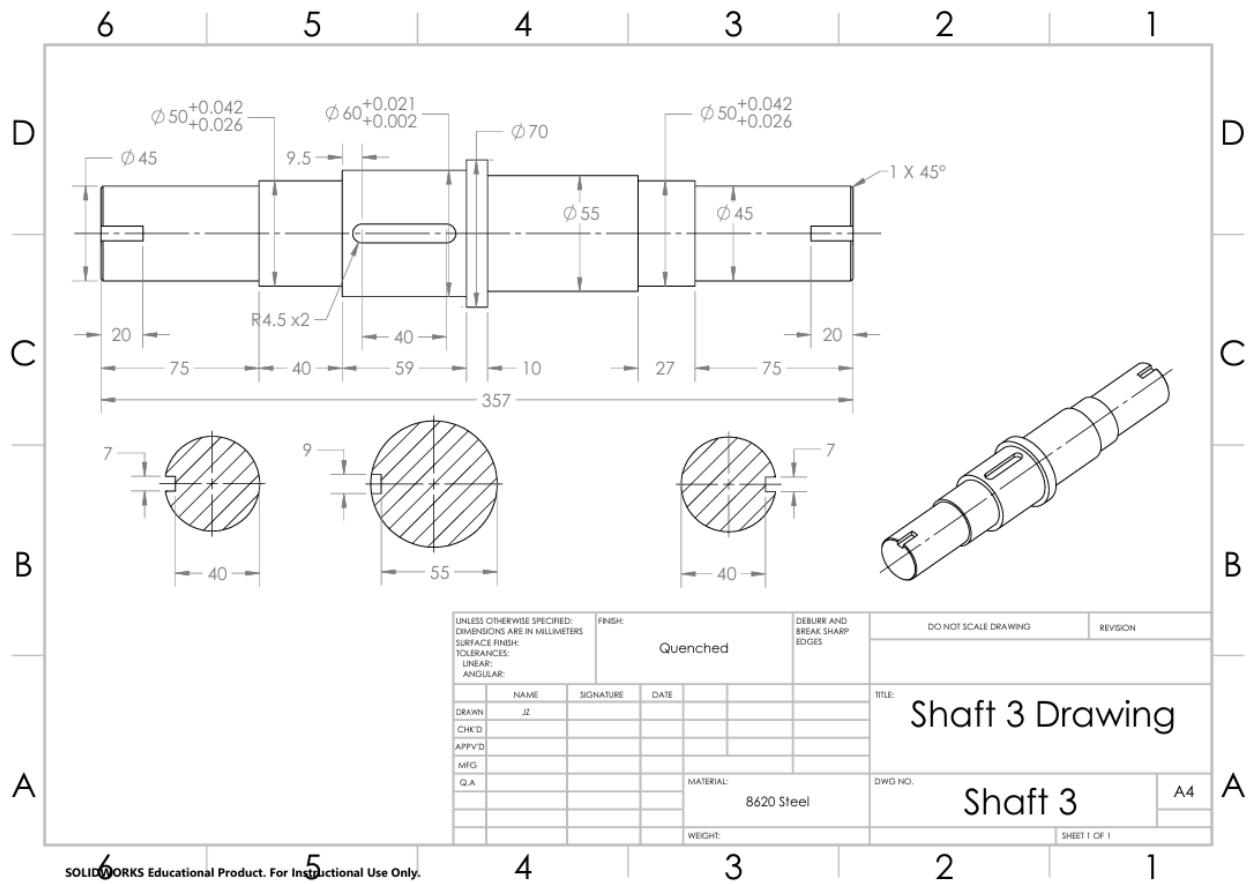


Figure 6: Shaft 3

Gear Part Drawings

Gear 1

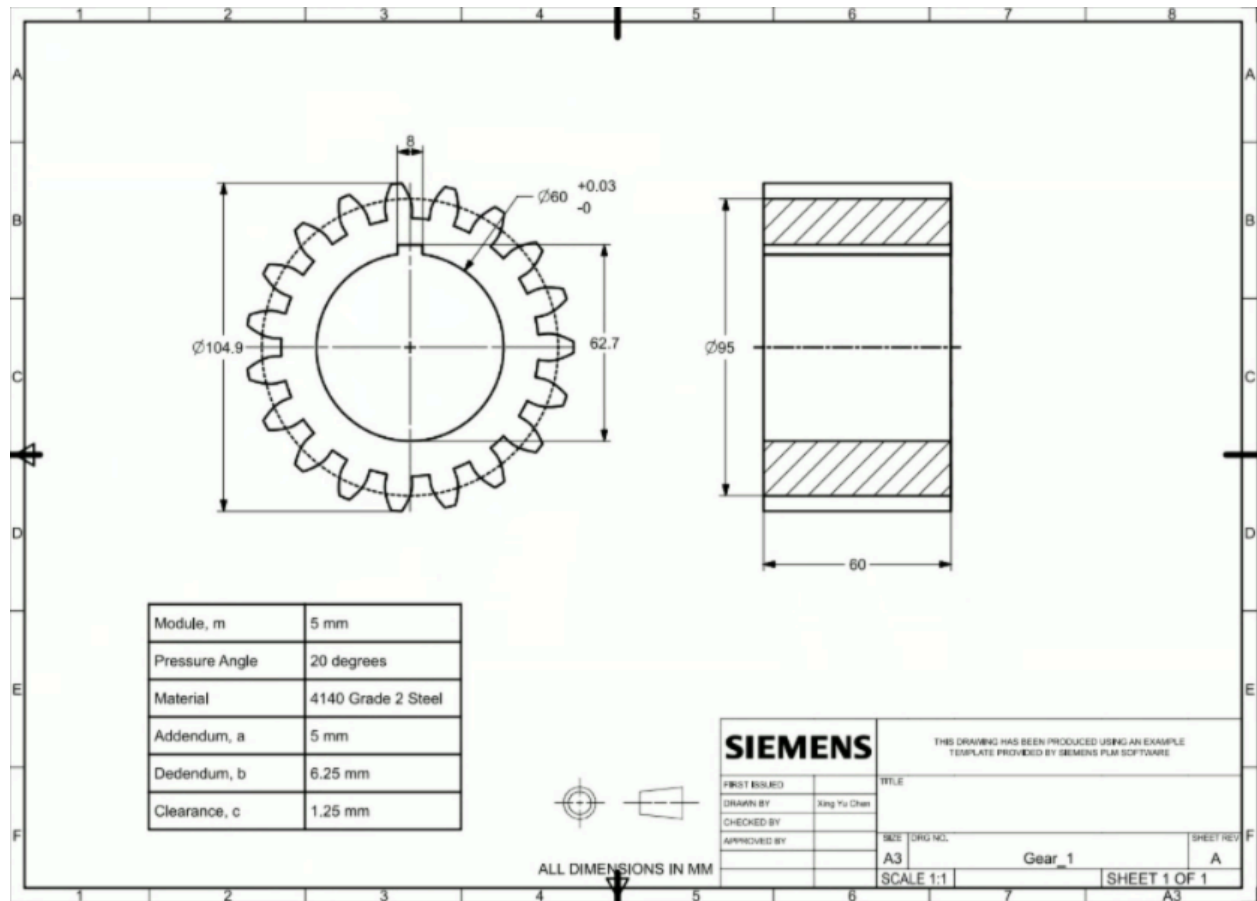


Figure 7: Gear 1

Gear 3

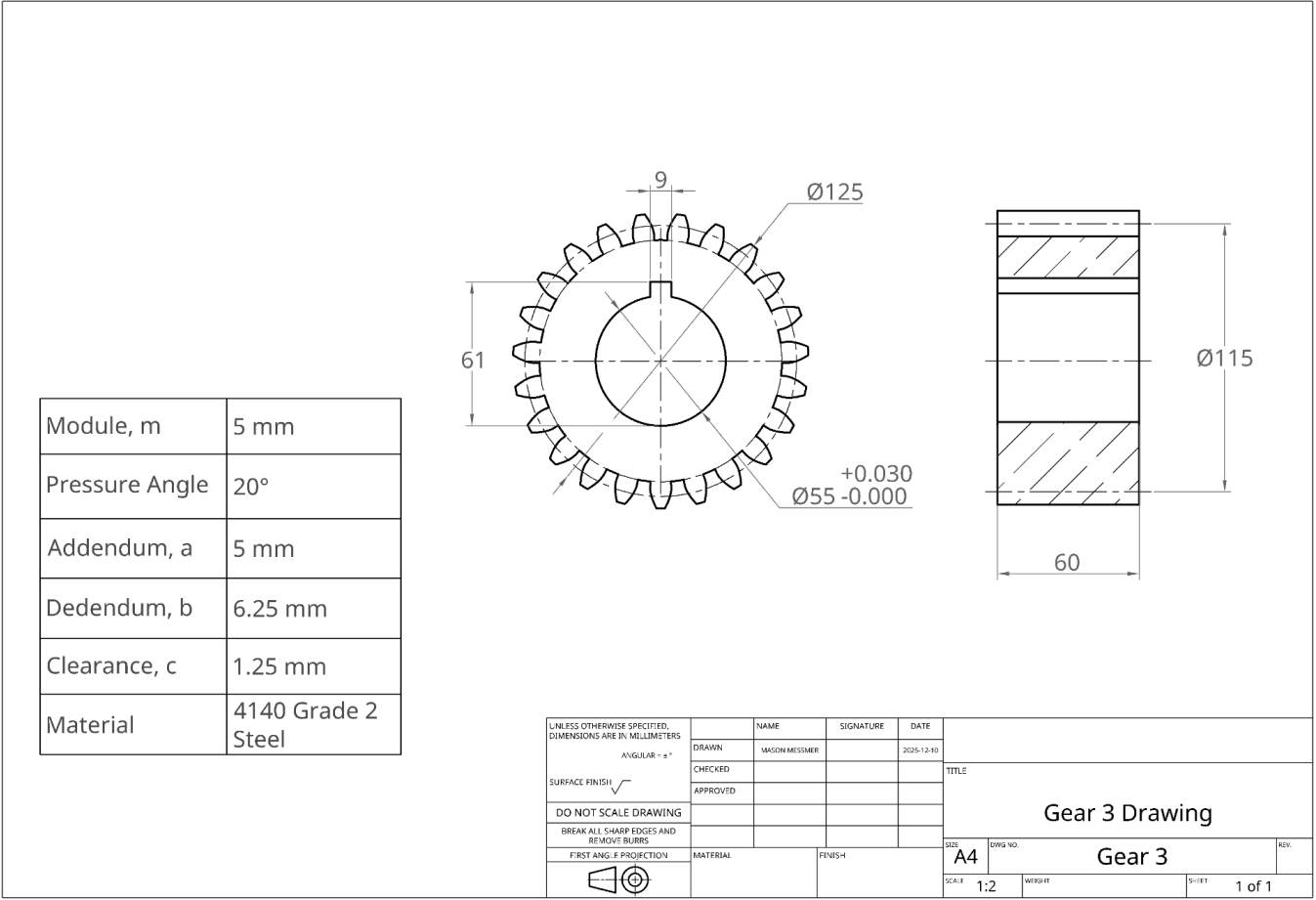


Figure 8: Gear 3

Gear 4

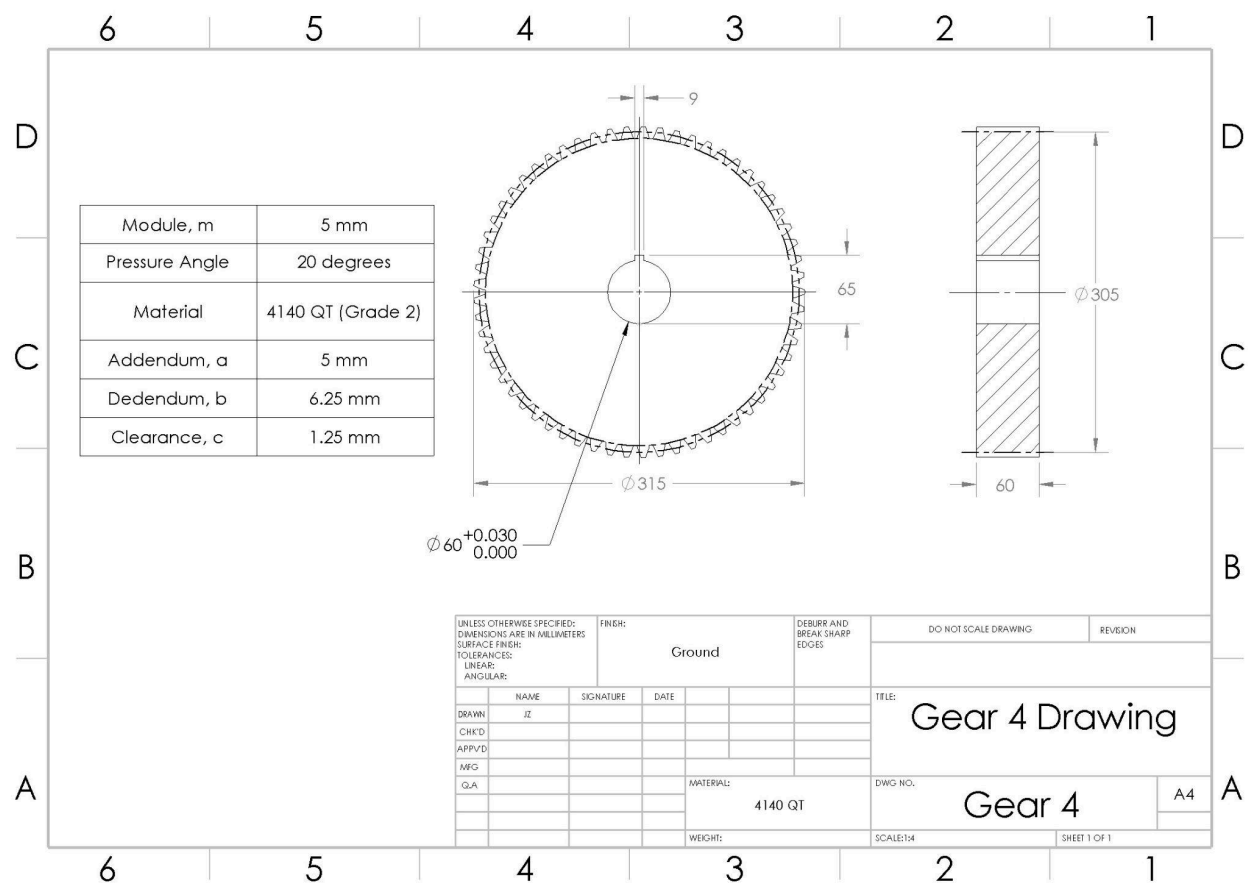


Figure 9: Gear 4

Appendix A

Shaft 1 Analysis

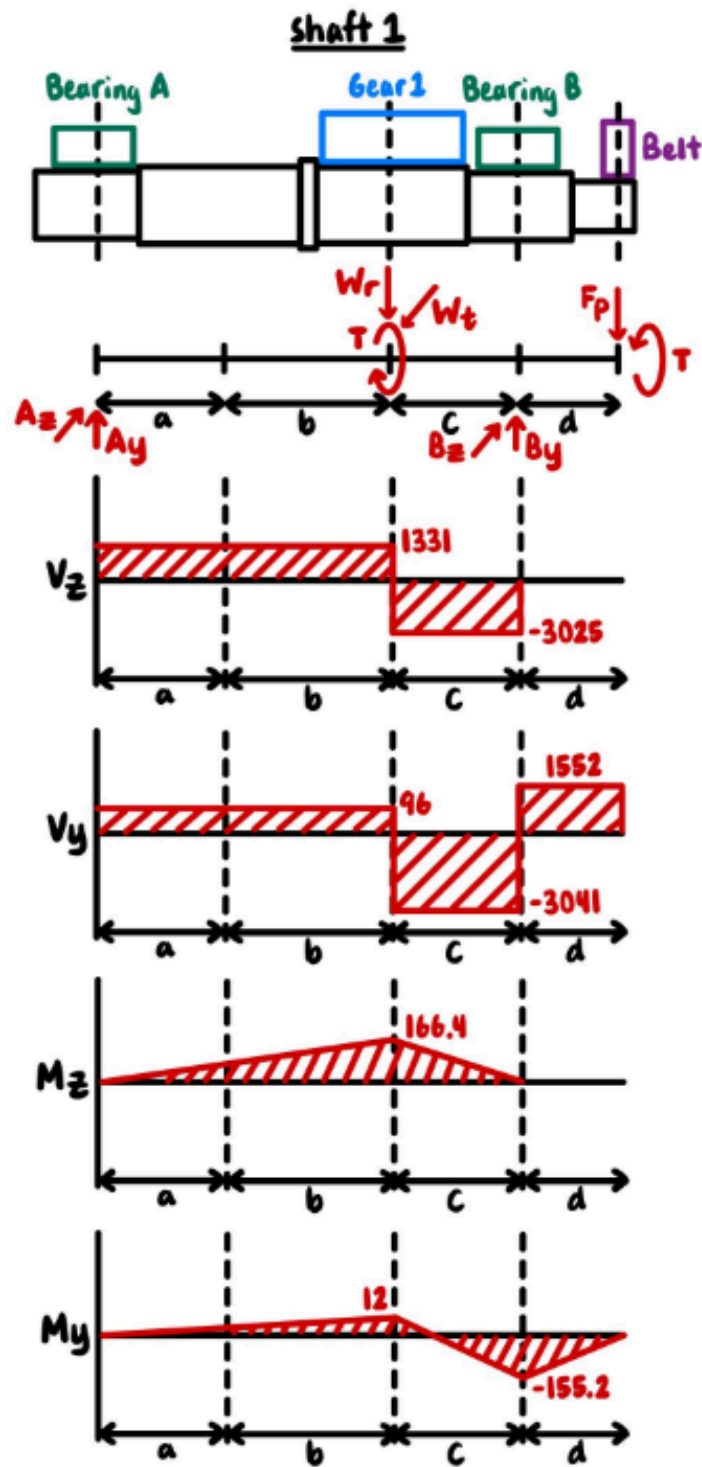


Figure 10: Shaft 1 Free Body and Bending Moment Diagrams

Given: $N_1 = 19$ $n_1 = 3000 \text{ rpm}$ $H = 65 \text{ kW}$ $m = 5 \text{ mm}$ $\phi = 20^\circ$ $D = 200 \text{ mm}$

Torque:
$$T_1 = \frac{60(H)}{2\pi n_1} = \frac{60(65000)}{2\pi(3000)} = 206.9 \text{ Nm}$$

Pitch Diameter of Gear 1: $d_1 = N_1 m = 19(5) = 95 \text{ mm} = 0.095 \text{ m}$

Forces on Gear 1:
$$W_1^t = \frac{2T_1}{d_1} = \frac{2(206.9)}{0.095} = 4356 \text{ N}$$

$$W_1^r = W_1^t \tan \phi = 4356 \tan(20^\circ) = 1585 \text{ N}$$

$$W_1^n = 0 \text{ N (spur gear)}$$

Input Pulley Calculations: $T_1 = \frac{D(F_1 - F_2)}{2} \rightarrow \frac{2T_1}{D} = \frac{2(206.9)}{0.2} = 2069 = F_1 - F_2$

Select V-belt: $\frac{F_1}{F_2} = 5 \rightarrow F_1 = 5F_2$, $2069 = 4F_2 \rightarrow F_2 = 517 \text{ N}$, $F_1 = 2586 \text{ N}$

Input Radial Force from Pulley: $F_p = \frac{F_1 + F_2}{2} = \frac{2586 + 517}{2} = 1552 \text{ N}$

Force and Moment Calculations: $a = c = 55 \text{ mm}$, $b = 70 \text{ mm}$, $d = 45 \text{ mm}$

$$\Sigma M_A^y = W_1^r(a + b) - B_y(a + b + c) + F_p(a + b + c + d) = 0$$

$$\Sigma M_A^y = 1585(0.125) - B_y(0.18) + 1552(0.225) = 0 \rightarrow B_y = 3041 \text{ N}$$

$$\Sigma F_y = A_y + B_y - W_1^r - F_p = 0 \rightarrow A_y = 1585 + 1552 - 3041 = 96 \text{ N}$$

$$\Sigma M_A^z = W_1^t(a + b) - B_z(a + b + c) = 0$$

$$\Sigma M_A^z = 4356(0.125) - B_z(0.18) = 0 \rightarrow B_z = 3025 \text{ N}$$

$$\Sigma F_z = A_z + B_z - W_1^t = 0 \rightarrow A_z = 4356 - 3025 = 1331 \text{ N}$$

Resultant Forces: $A_R = \sqrt{A_y^2 + A_z^2} = \sqrt{96^2 + 1331^2} = 1334 \text{ N}$

$$B_R = \sqrt{B_y^2 + B_z^2} = \sqrt{3041^2 + 3025^2} = 4289 \text{ N}$$

Most Critical Point: b to c, Critical Cross Section: (b to c)+ (i.e. to the right)

$$M_{max} = B_R(0.18 - 0.125) = 4289(0.055) = 236 \text{ Nm @ bearing B}$$

Minimum Shaft diameter (considering radial and torque): $\sigma_{allow} = 350 \text{ MPa (AISI 1045 Steel)}$

$$M = \sqrt{M_{max}^2 + 3T_1^2} = \sqrt{236^2 + 3(206.9)^2} = 429 \text{ Nm}$$

$$d_{min} = \sqrt[3]{\frac{32M}{\pi\sigma_{allow}}} = \sqrt[3]{\frac{32(429 \times 10^3)}{\pi(350 \times 10^6)}} = \sqrt[3]{0.012485} = 0.23 \text{ m} = 23 \text{ mm}$$

Correction for Keyway:

$$d_{corr} = 23(1.05) = 24 \text{ mm} \rightarrow \text{round up to 25 mm, adjusted to 40 mm for SF}$$

Critical Cross Section Infinite Life Calculations:

$$\sigma_a = \frac{32M_{max}}{\pi d^3} = \frac{32(236)}{\pi(0.04)^3} = 37.56 \text{ MPa}, \sigma_m = 0 \text{ (no axial force)}$$

$$\tau_m = \frac{1}{2}(\tau_{max}) = \frac{16T}{2\pi d^3} = \frac{8(206.9)}{\pi(0.04)^3} = 8.23 \text{ MPa} = \tau_a \text{ (on/off cycles)}$$

Safety Factor and Ultimate Tensile Strength: $SF = 2$, $S_{ut} = SF\sigma_{allow} = 2(350) = 700 \text{ MPa}$

Endurance Limit for $N > 10^6$: $S'_e = 0.5S_{ut} = 0.5(700) = 350 \text{ MPa}$ (for bending)

$$K_f = 3 \text{ (assumed)} \quad K_{fs} = 2.8 \text{ (assumed)} \quad k_t = 1 \quad k_m = 1$$

$$k_f = 1.58S_{ut}^{-0.085} = 1.58(700)^{-0.085} = 0.905 \text{ (ground finish)}$$

$$k_s = 1.189d^{-0.097} = 1.189(25)^{-0.097} = 0.92 \text{ (for bending)}$$

$$k_r = 0.82 \text{ (99\% reliability)}$$

$$S_e = k_f k_s k_r k_t k_m S'_e = (0.905)(0.92)(0.82)(1)(1)(350) = 239 \text{ MPa}$$

$$\text{Goodman Factor of Safety: } \frac{\sqrt{(K_f \sigma_a)^2 + 3(K_{fs} \tau_a)^2}}{S_e} + \frac{\sqrt{\sigma_m^2 + 3\tau_m^2}}{S_{ut}} = \frac{\sigma_{VM,a}}{S_e} + \frac{\sigma_{VM,m}}{S_{ut}} = \frac{1}{n_s}$$

$$\frac{1}{n_s} = \frac{\sqrt{(3(37.56 \times 10^6))^2 + 3(2.8(8.23 \times 10^6))^2}}{239 \times 10^6} + \frac{\sqrt{3(8.23 \times 10^6)^2}}{700 \times 10^6} = 0.521 \rightarrow n_s = \frac{1}{0.521} = 1.92 > 1.3$$

$$\text{Yield Factor of Safety: } n_y = \frac{\sigma_{allow}}{\sigma_{VM,a} + \sigma_{VM,m}} = \frac{350}{\sqrt{(37.56)^2 + 3(8.23)^2} + \sqrt{3(8.23)^2}} = 6.43$$

Satisfactory Goodman's factor of safety and von Mises yielding factor of safety.

Gear 1 Analysis

Given Parameters and Calculations from Shaft 1 Analysis:

$$N_1 = 19 \quad n_1 = 3000 \text{ rpm} \quad H = 65 \text{ kW} \quad m = 5 \text{ mm} \quad \phi = 20^\circ$$

$$T_1 = 206.9 \text{ Nm} \quad d_1 = 95 \text{ mm} \quad W_1^t = 4356 \text{ N} \quad W_1^r = 1585 \text{ N} \quad W_1^n = 0 \text{ N}$$

$$\text{Life Cycle Calculation: } h = 5 \text{ years} \times 52 \frac{\text{week}}{\text{year}} \times 7 \frac{\text{day}}{\text{week}} \times 8 \frac{\text{hour}}{\text{day}} = 14560 \text{ hours}$$

$$N_{\text{cycles}} = hn_1 = \frac{14560 \text{ hours} \times 60 \frac{\text{min}}{\text{hour}} \times 3000 \text{ rpm}}{10^6} = 2621 \text{ million cycles}$$

$$\text{Bending Stress Calculations: } HB = 450 \text{ (selected)}$$

$$\text{Stress Cycle Factor: } Y_N = 1.6831N_{\text{cycles}}^{-0.0323} = 1.31$$

$$\text{Allowable Bending Stress: } S_b = 0.703(450) + 113 = 429.4$$

$$K_T = 1 \text{ (assuming temperature } < 125^\circ\text{C)} \quad K_R = 1 \text{ (99\% reliability)}$$

$$\sigma_{b,allow} = \frac{S_b Y_N}{K_T K_R} = \frac{(429.4)(1.31)}{1(1)} = 562.5 \text{ MPa}$$

$$K_a = 1 \text{ (assuming no shock)} \quad b_w = 12m = 12(5) = 60 \text{ mm}$$

$$C_{mc} = 1 \text{ (uncrowned teeth)} \quad C_{pf} = \frac{b_w}{10d_1} - 0.0375 + 0.000492b_w = 0.02569$$

$$C_{pm} = 1.1 \text{ (for } \frac{S_1}{S} = \frac{a+b-0.5(a+b+c)}{a+b+c} = \frac{0.125-0.5(0.18)}{0.18} = \frac{0.035}{0.18} \approx 0.194 > 0.175)$$

$$\text{For Commercial Gears: } A = 0.127B = 6.22 \times 10^{-4} \quad C = -1.69 \times 10^{-7}$$

$$C_{ma} = A + Bb_w + Cb_w^2 = 0.127 + 6.22 \times 10^{-4}(60) - 1.69 \times 10^{-7}(60)^2 = 0.1637$$

$$C_e = 1 \text{ (assuming no adjustment)}$$

$$K_m = 1 + C_{mc}(C_{pf}C_{pm} + C_{ma}C_e) = 1 + 1(0.02569(1.1) + 0.1637(1)) = 1.192$$

$$K_s = 1 \text{ (size factor)} \quad K_i = 1 \text{ (not idler)} \quad K_b = 1 \text{ (Rim thickness)}$$

$$V = \frac{\pi n_1 d_1}{60} = \frac{\pi(3000)(0.095)}{60} = 14.923$$

$$Q_v = 6 \text{ (for high quality commercial gears)}$$

$$B = \frac{1}{4} (12 - Q_v)^{2/3} = 0.825 \quad A = 50 + 56(1 - B) = 59.773$$

$$K_v = \left(\frac{A + \sqrt{200V}}{A} \right)^B = \left(\frac{59.773 + \sqrt{200(14.923)}}{59.773} \right)^{0.825} = 1.71$$

$$Y_j \approx 0.32 (N_1 = 19)$$

$$\sigma_b = \frac{W^t P}{b_w Y_j} K_a K_s K_m K_v K_t K_b = \frac{4356(\frac{1}{5})}{60(0.32)} (1)(1)(1.192)(1.71)(1)(1) = 92.49 \text{ MPa}$$

$$n_b = \frac{\sigma_{b,allow}}{\sigma_b} = \frac{562.5}{92.49} = 6.08$$

This gear design has a satisfactory bending stress factor of safety (greater than 1.1).

Contact Stress Calculations:

$$HB_{pinion} = 450 \quad HB_{gear} = 375 \quad K_T = 1 \quad K_R = 1 (99\% \text{ reliability})$$

$$Z_N = 2.466 N_{cycles}^{-0.056} = 1.59 \quad C_H = 1 (for \text{ pinion})$$

$$S_{c, grade 2} = 2.41 HB_{pinion} = 2.41(450) + 237 = 1321.5 \text{ MPa}$$

$$\sigma_{c,allow} = \frac{S_c Z_N C_H}{K_T K_R} = \frac{(1321.5)(1.59)(1)}{(1)(1)} = 2101.2 \text{ MPa}$$

$$K_a = 1 \quad K_s = 1 \quad K_m = 1.192 \quad K_v = 1.71$$

$$E_g \approx 200 \text{ GPa} \quad \nu_g \approx 0.285 \quad E_p \approx 200 \text{ GPa} \quad \nu_p \approx 0.285$$

$$K_E = \sqrt{\frac{2}{\frac{1-\nu_p^2}{E_p} + \frac{1-\nu_g^2}{E_g}}} = 466563 \quad I = \frac{\pi \cos(20^\circ) \sin(20^\circ)}{1 + \frac{95}{295}} = 0.764$$

$$\sigma_c = K_E \left(\frac{W_1^t}{b_w d_1} \frac{1}{I} K_a K_s K_m K_v \right)^{1/2} = 466563 \sqrt{\frac{4356}{60(95)} \left(\frac{1}{0.764} \right) (1)(1)(1.192)(1.71)} = 666.2 \text{ MPa}$$

$$n_s = \frac{\sigma_{c,allow}}{\sigma_c} = \frac{2101.2}{666.2} = 3.15$$

This gear design has a satisfactory contact stress factor of safety (greater than 1.1).

Material selection could be modified (for weaker HB) since both factors of safety are well above 1.1 (slightly over-engineered), but for the purposes of this project and maintaining consistency with the other gears, this choice of material is sufficient and justified.

Bearings A and B Analysis

$$A_p = 1 \quad n_1 = 3000 \text{ rpm} \quad m = 3 \quad P_r = B_R = 4289 \text{ N} \quad P_a = A_R = 1334 \text{ N}$$

$$L_{10} = N_{\text{cycles}} = 2621 \text{ million cycles} \quad P = P_r = 4289 \text{ N (initial selection)}$$

$$C = PL_{10}^{\frac{1}{m}} = 4289(2621)^{\frac{1}{3}} = 59135 \text{ N} \rightarrow \text{Select 6310 SKF Deep Groove Ball Bearings}$$

$$d = 50 \text{ mm} \quad OD = 110 \text{ mm} \quad W = 27 \text{ mm} \quad C = 61800 \text{ N} \quad C_o = 38000 \text{ N}$$

$$\frac{P_a}{C_o} = \frac{1334}{38000} = 0.035 \rightarrow 0.025 < \frac{P_a}{C_o} < 0.04 \rightarrow 0.22 < e < 0.24$$

$$\text{Linear Interpolation: } e \approx 0.22 + \frac{0.035 - 0.025}{0.04 - 0.025} \times (0.024 - 0.022) = 0.23 \text{ (rounded)}$$

$$\frac{P_a}{P_r} = \frac{1334}{4289} = 0.31 \rightarrow \frac{P_a}{P_r} > e \rightarrow X = 0.56, 1.8 < Y < 2.0$$

$$\text{Linear Interpolation: } Y \approx 2.0 + \frac{0.035 - 0.025}{0.04 - 0.025} \times (1.8 - 2.0) = 1.87 \text{ (rounded)}$$

$$P = A_p(XP_r + YP_a) = (1)(0.56(4289) + 1.87(1334)) = 4896 \text{ N}$$

$$C = PL_{10}^{\frac{1}{m}} = 4896(2621)^{\frac{1}{3}} = 67504 \text{ N} > 61800 \text{ N (reselect bearings: SKF 6410)}$$

$$d = 50 \text{ mm} \quad OD = 130 \text{ mm} \quad W = 31 \text{ mm} \quad C = 87100 \text{ N} \quad C_o = 52000 \text{ N}$$

$$\frac{P_a}{C_o} = \frac{1334}{52000} = 0.025 \text{ (rounded)} \rightarrow e = 0.22 < \frac{P_a}{C_o} \rightarrow X = 0.56, Y = 2.0$$

$$P = A_p(XP_r + YP_a) = (1)(0.56(4289) + 2.0(1334)) = 5070 \text{ N}$$

$$C = PL_{10}^{\frac{1}{m}} = 5070(2621)^{\frac{1}{3}} = 69903 \text{ N} < 87100 \text{ N (keep SKF 6410)}$$

Select SKF 6410 for both bearing A and B for simplicity, ease of assembly, and maintenance. Bearing A has less applied load, so this calculation for bearing B is sufficient.

Selected SKF 6410: ID = 50 mm OD = 130 mm W = 31 mm

Appendix B

Given and predetermined values

$$n_1 = 3000 \text{ rpm} \quad H = 65 \text{ kW} \quad N_4 = 61 \quad m = 5 \text{ mm} \quad \phi = 20^\circ$$

$$N_1 = 19 \quad N_2 = 59 \quad N_3 = 23$$

Calculating shaft 2 rpm and torque

$$\frac{n_1}{n_2} = \frac{N_2}{N_1} \Rightarrow n_2 = \frac{(19)}{(59)} (3000) = 966.1 \text{ rpm}$$

$$T_2 = \frac{(60)(65,000)}{2\pi(966.1)} = 642.5 \text{ Nm}$$

Calculating forces on gears 2 & 3

$$d_2 = 59(5) = 295 \text{ mm} \quad d_3 = 23(5) = 115 \text{ mm}$$

$$W_2^t = 642.5 / (0.295/2) = 4356 \text{ N} \quad W_3^t = 642.5 / (0.115/2) = 11174 \text{ N}$$

$$W_2 = \frac{4356}{\cos(20)} = 4636 \text{ N} \quad W_3 = \frac{11174}{\cos(20)} = 11891 \text{ N}$$

$$W_2^r = 4356 \tan(20) = 1585 \text{ N} \quad W_3^r = 11174 \tan(20) = 4067 \text{ N}$$

Shaft 2 Analysis

Force and moment calculations

$$a = c = 55 \text{ mm} \quad b = 70 \text{ mm}$$

$$\Sigma M_C^y = 1585(0.125) + Dz(0.18) - 4067(0.055) = 0$$

$$\Sigma M_C^z = Dy(0.18) - 11174(0.055) - 4356(0.125) = 0$$

$$Dz = 141.7 \text{ N} \quad Cz = 4067 - 1585 - 141.7 = 2340 \text{ N}$$

$$Dy = 6439 \text{ N} \quad Cy = 11174 + 4356 - 6439 = 9090 \text{ N}$$

$$D_R = \sqrt{141.7^2 + 6439^2} = 6441 \text{ N} \quad C_R = \sqrt{2340^2 + 9090^2} = 9387 \text{ N}$$

$$M_{max} = 516.3 \text{ Nm @ gear 3}$$

Minimum shaft diameter

$$\tau_{allow} = \frac{0.557(360 \text{ MPa})}{1.3} = 160 \text{ MPa (8620 steel, SF of 1.3)} \quad d_{min} = \sqrt[3]{\frac{16T}{\pi\tau_{allow}}} = 27 \text{ mm}$$

Correction for keyway

$$d_{corr} = 27(1.05) = 30 \text{ mm} \Rightarrow \text{sizing up to 55 mm @ gear 3 (50mm else) for better SF}$$

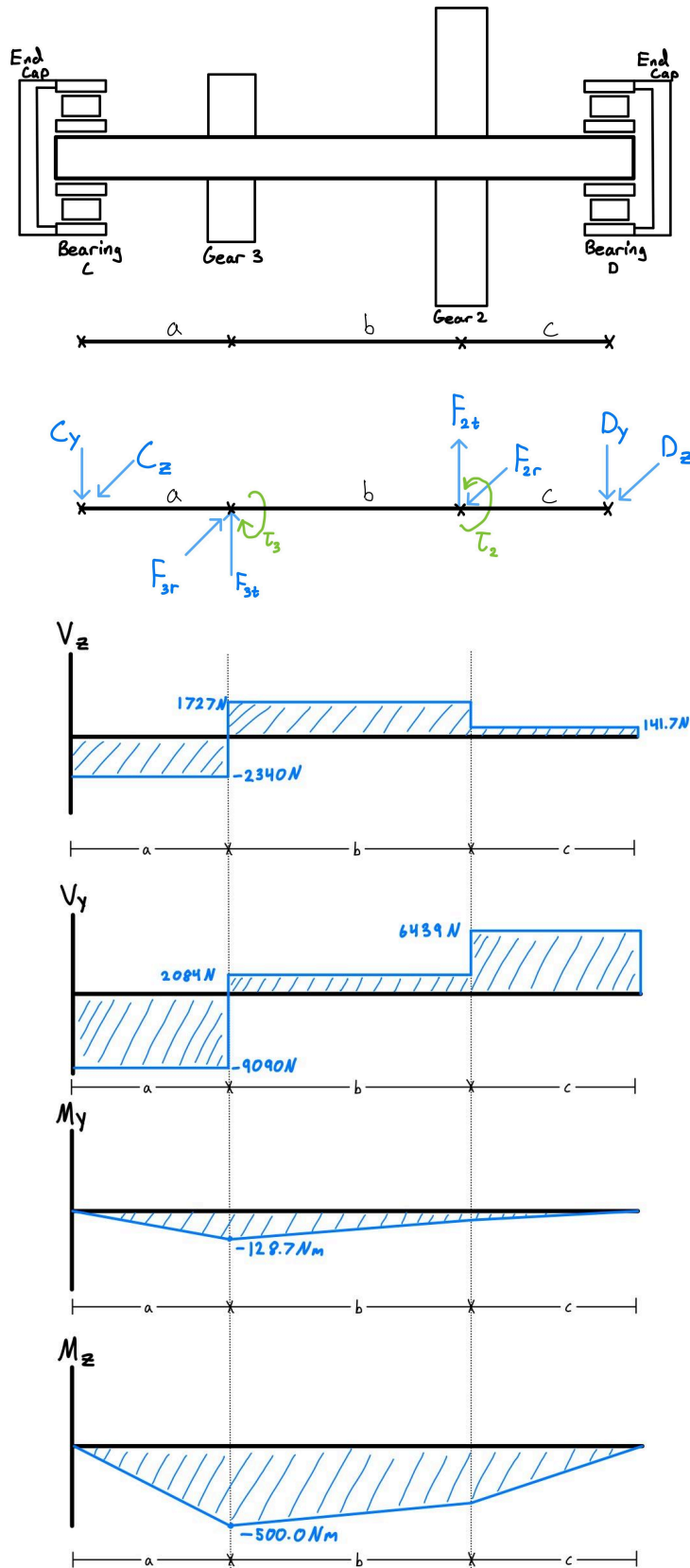


Figure 11: Shaft 2 Assembly, Free Body, Shear, and Moment Diagrams

Critical cross section infinite life calculations

$$\sigma_a = \frac{32M}{\pi d^3} = 31.6 \text{ MPa}$$

$$\tau_m = \frac{8T}{\pi d^3} = 9.8 \text{ MPa} = \tau_a$$

$$K_f = 3; K_{fs} = 2.8 \text{ (assumed)} \quad S_{ut} = 635 \text{ MPa} \quad S_e' = 0.5 S_{ut} = 317.5 \text{ MPa}$$

$$k_f = 1.58 S_{ut}^{-0.085} = 0.912896 \quad k_s = 1.189 d^{-0.112} = 0.759034$$

$$k_r = 0.82 \text{ (99\% reliability)} \quad k_t = 1 \quad k_m = 1$$

$$\text{Goodman's Line: } \frac{\sqrt{3(31.6)^2 + 3[2.8(9.8)]^2}}{(0.912896)(0.759034)(0.82)(317.5)} + \frac{\sqrt{3}(9.8)}{635} = \frac{1}{n_s} \Rightarrow n_s = 1.63$$

$$\text{Yielding Line: } n_y = \frac{360}{\sqrt{(31.6)^2 + 3(9.8)^2} + \sqrt{3}(9.8)} = 6.80$$

This shaft design has a satisfactory Goodman's factor of safety as well as von Mises yielding factor of safety. (greater than 1.27)

Gear 2 Analysis

Contact Ratio calculation

$$C_r = \frac{1}{p_c \cos(\phi)} [\sqrt{r_{op}^2 - r_{bp}^2} + \sqrt{r_{og}^2 - r_{bg}^2}] - \frac{c_d \tan(\phi)}{p_c}$$

$$p_c = \pi(0.005)m \quad c_d = 0.195m \quad r_o = \frac{1}{2}(d_c + 2m) \quad r_b = \frac{1}{2}(d_c - 2.5m)$$

$$r_{op} = 0.0525m \quad r_{bp} = 0.04125m \quad r_{og} = 0.1525m \quad r_{bg} = 0.14125m$$

$$C_r = 1.576$$

Shaft cycle calculation

$$\text{Expected life} = 5\text{yrs} \times 52 \frac{\text{week}}{\text{yr}} \times 7 \frac{\text{day}}{\text{week}} \times 8 \frac{\text{hr}}{\text{day}} = 14560 \text{ hrs}$$

$$N_{\text{cycles}} = 14560 \text{ hrs} \times 60 \frac{\text{min}}{\text{hr}} \times 966.1 \text{ rpm} = 844.0 \text{ million cycles}$$

Bending stress calculation

$$Y_n = 1.6831 N_{\text{cycles}}^{-0.0323} = 0.867 \quad HB = 375$$

$$S_{b, \text{grade 1}} = 0.533(375) + 88.3 = 288.2 \text{ MPa}$$

$$k_t = 1 \text{ (assuming temperature } < 125^\circ \text{C)} \quad k_r = 1$$

$$\sigma_{b, \text{allow}} = \frac{(288.2)(0.867)}{1(1)} = 249.7 \text{ MPa}$$

$$Y_j \approx 0.41 \text{ (} N = 59 \text{)} \quad K_a = 1 \text{ (assuming no shock)}$$

$$Cmc = 1 \quad Cpf = \frac{bw}{10d} - 0.0375 + 0.000492bw = 0.055178$$

$$Cpm = 1 \text{ (for } \frac{S_1}{S} \approx 0.156 < 0.175) \quad Ce = 1 \text{ (assuming no adjustment)}$$

$$Cma = 0.127 + 0.000622(60) - 1.69 \times 10^{-7}(60)^2 = 0.1637 \text{ (for commercial gears)}$$

$$k_m = 1 + 1(0.055178 + 0.1637) = 1.218792$$

$$k_s = 1 \quad k_a = 1 \quad k_i = 1 \quad k_b = 1$$

$$V = \frac{\pi nd}{60} = 14.92257 \quad Q_v = 6 \text{ (for high quality commercial gears)}$$

$$B = \frac{(12-Q_v)^{\frac{2}{3}}}{4} = 0.825 \quad A = 50 + 56(1 - B) = 59.773$$

$$K_v = \left(\frac{A + \sqrt{200V}}{A} \right)^B = 1.71$$

$$\sigma_b = \frac{4356}{(0.005(0.060)(0.41))} (1)(1)(1.218792)(1.71)(1)(1) = 73.8 \text{ MPa}$$

$$n_b = \frac{249.7}{73.8} = 3.39$$

This gear design has a satisfactory bending stress factor of safety (greater than 1.1).

Contact stress calculation

$$HB_{pinion} = 450 \quad HB_{gear} = 375 \quad K_t = 1 \quad K_r = 1 \text{ (99\% reliability)}$$

$$Z_n = 2.466N_{cycles}^{-0.056} = 0.780$$

$$C_H = 1 + [(0.00889)\left(\frac{450}{375}\right) - 0.00829]\left(\frac{59}{19} - 1\right) = 1.0050$$

$$S_{c, grade 1} = 2.22(450) + 200 = 1321.5$$

$$\sigma_{c, allow} = \frac{S_c Z_n C_H}{K_t K_r} = 809.4 \text{ MPa}$$

$$K_a = 1 \quad K_s = 1 \quad K_m = 1.218792 \quad K_v = 1.71$$

$$E_g \approx 200 \text{ GPa} \quad \nu_g \approx 0.285 \quad E_p \approx 200 \text{ GPa} \quad \nu_p \approx 0.285$$

$$K_E = \sqrt{\frac{2}{\frac{1-\nu_p^2}{E_p} + \frac{1-\nu_g^2}{E_g}}} = 466563 \quad I = \frac{\pi \cos(20) \sin(20)}{1 + \frac{95}{295}} = 0.7637$$

$$\sigma_c = 466563 \sqrt{\frac{4356}{(0.06)(0.095)(0.7637)}} (1)(1)(1.218792)(1.71) = 673.5 \text{ MPa}$$

$$n_s = \frac{809.4}{673.5} = 1.20$$

This gear design has a satisfactory contact stress factor of safety (greater than 1.1).

Gear 3 Analysis

Contact Ratio calculation

$$C_r = \frac{1}{p_c \cos(\phi)} [\sqrt{r_{op}^2 - r_{bp}^2} + \sqrt{r_{og}^2 - r_{bg}^2}] - \frac{c_d \tan(\phi)}{p_c}$$

$$p_c = \pi(0.005)m \quad c_d = 0.21m \quad r_o = \frac{1}{2}(d_c + 2m) \quad r_b = \frac{1}{2}(d_c - 2.5m)$$

$$r_{op} = 0.0652m \quad r_{bp} = 0.05125m \quad r_{og} = 0.1575m \quad r_{bg} = 0.14625m$$

$$C_r = 1.518$$

Bending stress calculation

$$Y_n = 1.6831N_{cycles}^{-0.0323} = 0.867 \quad HB = 450$$

$$S_{b, grade 2} = 0.703(450) + 113 = 429.4$$

$$k_t = 1 \text{ (assuming temperature } < 125^\circ C) \quad k_r = 1$$

$$\sigma_{b, allow} = \frac{(429.4)(0.867)}{1(1)} = 372.0 \text{ MPa}$$

$$Y_j \approx 0.34 (N = 23) \quad K_a = 1 \text{ (assuming no shock)}$$

$$Cmc = 1 \quad Cpf = \frac{bw}{10d} - 0.0375 + 0.000492bw = 0.044194$$

$$Cpm = 1 \text{ (for } \frac{S_1}{S} \approx 0.156 < 0.175) \quad Ce = 1 \text{ (assuming no adjustment)}$$

$$Cma = 0.127 + 0.000622(60) - 1.69 \times 10^{-7}(60)^2 = 0.1637 \text{ (for commercial gears)}$$

$$k_m = 1 + 1(0.044194 + 0.1637) = 1.207808$$

$$k_s = 1 \quad k_a = 1 \quad k_i = 1 \quad k_b = 1$$

$$V = \frac{\pi nd}{60} = 5.81778Q_v = 6 \text{ (for high quality commercial gears)}$$

$$B = \frac{(12-Q_v)^{\frac{2}{3}}}{4} = 0.825 \quad A = 50 + 56(1 - B) = 59.773$$

$$K_v = \left(\frac{A + \sqrt{200V}}{A} \right)^B = 1.45$$

$$\sigma_b = \frac{11174}{(0.005)(0.060)(0.34)} (1)(1)(1.207808)(1.45)(1)(1) = 192.1 \text{ MPa}$$

$$n_b = \frac{372.0}{192.1} = 1.94$$

This gear design has a satisfactory bending stress factor of safety (greater than 1.1).

Contact stress calculation

$$HB_{pinion} = 450 \quad HB_{gear} = 375 \quad K_t = 1 \quad K_r = 1 \text{ (99\% reliability)}$$

$$Zn = 2.466N_{cycles}^{-0.056} = 0.780$$

$$C_H = 1 \text{ for pinion}$$

$$S_{c, grade 2} = 2.41(450) + 237 = 1321.5$$

$$\sigma_{c,allow} = \frac{S_c Z C_H}{K_t K_r} = 1030.8 \text{ MPa}$$

$$K_a = 1 \quad K_s = 1 \quad K_m = 1.207808 \quad K_v = 1.45$$

$$E_g \approx 200 \text{ GPa} \quad \nu_g \approx 0.285 \quad E_p \approx 200 \text{ GPa} \quad \nu_p \approx 0.285$$

$$K_E = \sqrt{\frac{2}{\frac{1-\nu_p^2}{E_p} + \frac{1-\nu_g^2}{E_g}}} = 466563 \quad I = \frac{\pi \cos(20) \sin(20)}{1 + \frac{115}{305}} = 0.7332$$

$$\sigma_c = 466563 \sqrt{\frac{11174}{(0.06)(0.115)(0.7332)}} (1)(1)(1.207808)(1.45) = 918.1 \text{ MPa}$$

$$n_s = \frac{1030.8}{918.1} = 1.12$$

This gear design has a satisfactory contact stress factor of safety (greater than 1.1).

Bearings C and D Analysis

$$n_4 = 966.1 \text{ rpm} \quad m = 10/3 \text{ (for roller bearings)}$$

$$C_R = 9387 \text{ N} \quad D_R = 6441 \text{ N} \quad C_R > D_R \Rightarrow P_r = 9387 \text{ N}$$

$$A_P = 1 \quad X = 1 \quad P_a = 0 \quad P = P_r X = 9387 \text{ N}$$

$$L_{10} = 844 \text{ million cycles}$$

$$C_{calc} = 9387(844)^{\frac{3}{10}} = 70862 \text{ N}$$

$$L_{expected} = \left(\frac{C}{P_r}\right)^{\frac{3}{10}} = \left(\frac{73500}{9387}\right)^{\frac{10}{3}} = 953.3 \text{ million cycles}$$

From the bearing catalog: bearing SKF NU 210 ECM, ID of 50 mm, OD of 90 mm, width of 20 mm and a specified load rating of 73500 N satisfies our needs for bearing C's radial load.

As bearing D has a lower radial load, for simplicity, ease of assembly, and maintenance, we will choose SKF NU 210 ECM for both bearings.

Appendix C

Given and predetermined values

$$\begin{array}{lllll} n_1 = 3000 \text{ rpm} & H = 65 \text{ kW} & N_4 = 61 & m = 5 \text{ mm} & \phi = 20^\circ \\ & N_1 = 19 & N_2 = 59 & N_3 = 23 & \end{array}$$

Calculating shaft 3 rpm

$$\frac{n_1}{n_4} = \frac{N_2 N_4}{N_1 N_3} \Rightarrow n_4 = \frac{(19)(23)}{(59)(61)} (3000) = 364.3 \text{ rpm}$$

Calculating torque and forces on gear 4

$$\begin{aligned} T_4 &= \frac{(60)(65,000)}{2\pi(364.3)} = 1704 \text{ Nm} \\ d &= 61(5) = 305 \text{ mm} \\ W_4^t &= 1704/(305/2) = 11174 \text{ N} \\ W_4 &= \frac{11174}{\cos(20)} = 11891 \text{ N} \\ W_4^r &= 11174 \tan(20) = 4067 \text{ N} \end{aligned}$$

Shaft 3 Analysis

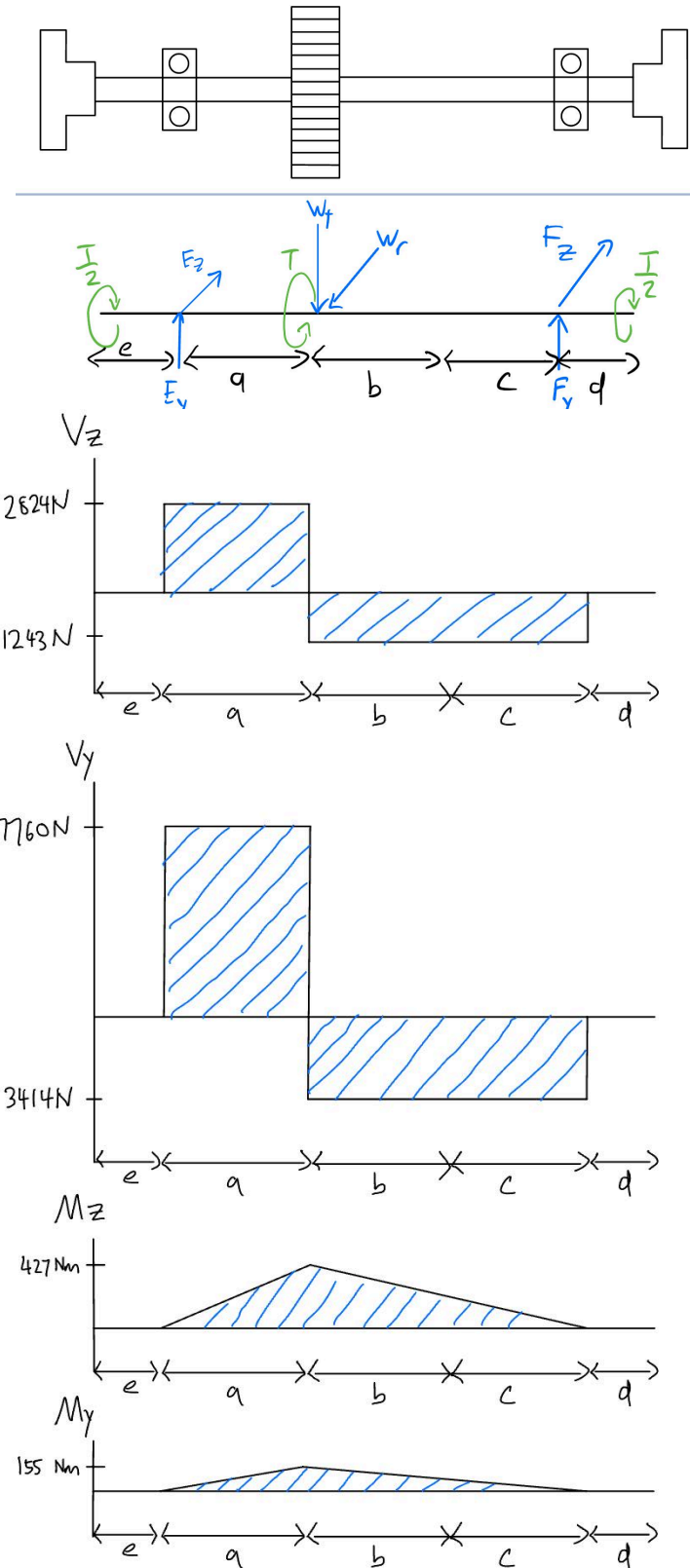


Figure 12: Shaft 3 Drawing, FBD, Shear and Moment Diagrams

Force and moment calculations

$$a = c = 55\text{mm} \quad b = 70\text{mm}$$

$$\Sigma M_E^y = 11174(0.055) - Fy(0.18) = 0$$

$$Fy = 3414 \text{ N}$$

$$Ey = 11174 - 3414 = 7760 \text{ N}$$

$$\Sigma M_E^z = 4067(0.055) - Fz(0.18) = 0$$

$$Fz = 1243 \text{ N}$$

$$Ez = 4067 - 1243 = 2824 \text{ N}$$

$$E_R = \sqrt{7760^2 + 2824^2} = 8258 \text{ N}$$

$$F_R = \sqrt{3414^2 + 1243^2} = 3633 \text{ N}$$

$$M_{max} = 454 \text{ Nm @ gear 4}$$

Minimum shaft diameter

$$\tau_{allow} = \frac{0.557(385\text{MPa})}{2} = 107 \text{ MPa (8620 steel, SF of 2)}$$

$$d_{min} = \sqrt[3]{\frac{16T}{\pi\tau_{allow}}} = 43 \text{ mm}$$

Correction for keyway

$$d_{corr} = 43(1.05) = 45 \text{ mm} \Rightarrow \text{sizing up to 60 mm for a better SF}$$

Critical cross section infinite life calculations

$$\sigma_a = \frac{32M}{\pi d^3} = 21.4 \text{ MPa}$$

$$\tau_m = \frac{8T}{\pi d^3} = 20.1 \text{ MPa} = \tau_a$$

$$K_f = 3 \text{ (assumed)} \quad K_{fs} = 2.8 \text{ (assumed)} \quad S_{ut} = 635 \text{ MPa} \quad S_e' = 317.5 \text{ MPa}$$

$$k_f = 1.58S_{ut}^{-0.085} = 0.912896 \quad k_s = 1.189d^{-0.112} = 0.75$$

$$k_r = 0.82 \text{ (99\% reliability)} \quad k_f = 1 \quad k_m = 1$$

$$\frac{\sqrt{3(21.4)^2 + 3[2.8(20.1)]^2}}{(0.912896)(0.75)(0.82)(317.5)} + \frac{\sqrt{3}(20.1)}{635} = \frac{1}{n_s} \Rightarrow n_s = 1.56$$

$$n_y = \frac{385}{\sqrt{21.4^2 + 3(40.2)^2}} = 5.3$$

This shaft design has a satisfactory Goodman's factor of safety as well as von Mises yielding factor of safety.

Gear 4 Analysis

Shaft cycle calculation

$$\text{Expected life} = 5\text{yrs} \times 52 \frac{\text{week}}{\text{yr}} \times 7 \frac{\text{day}}{\text{week}} \times 8 \frac{\text{hr}}{\text{day}} = 14560 \text{ hrs}$$

$$N_{\text{cycles}} = 14560 \text{ hrs} \times 60 \frac{\text{min}}{\text{hr}} \times 364.3 \text{ rpm} = 318 \text{ million cycles}$$

Bending stress calculation

$$Y_n = 1.6831 N_{\text{cycles}}^{-0.0323} = 0.894 \quad HB = 375$$

$$S_{b, \text{Grade 2}} = 0.703(375) + 113 = 377$$

$$k_t = 1 \text{ (assuming temperature } < 125^\circ \text{C)} \quad k_r = 1$$

$$\sigma_{b, \text{allow}} = \frac{(248.2)(0.894)}{1(1)} = 258 \text{ MPa}$$

$$Y_j \approx 0.42 \text{ (} N = 61 \text{)} \quad K_a = 1 \text{ (assuming no shock)}$$

$$Cmc = 1 \quad Cpf = \frac{bw}{10d} - 0.0375 + 0.000492bw = -0.017857$$

$$Cpm = 1 \text{ (for } \frac{S_1}{S} \approx 0.156 < 0.175 \text{)} \quad Ce = 1 \text{ (assuming no adjustment)}$$

$$Cma = 0.127 + 0.000622(60) - 1.69 \times 10^{-7}(60)^2 = 0.1637 \text{ (for commercial gears)}$$

$$k_m = 1 + 1(-0.017857 + 0.1637) = 1.145843$$

$$k_s = 1 \quad k_a = 1 \quad k_i = 1 \quad k_b = 1$$

$$V = \frac{\pi nd}{60} = 5.81778 Q_v = 6 \text{ (for high quality commercial gears)}$$

$$B = \frac{(12 - Q_v)^{\frac{2}{3}}}{4} = 0.825 \quad A = 50 + 56(1 - B) = 59.773$$

$$K_v = \left(\frac{A + \sqrt{200V}}{A} \right)^B = 1.45$$

$$\sigma_b = \frac{11174}{(0.005(0.060)(0.42))} (1)(1)(1.145843)(1.45)(1)(1) = 147.48 \text{ MPa}$$

$$n_b = \frac{377}{147.48} = 2.5$$

This gear design has a satisfactory bending stress factor of safety (greater than 1.1).

Contact stress calculations

$$HB_{\text{pinion}} = 450 \quad HB_{\text{gear}} = 375 \quad K_t = 1 \quad K_r = 1 \text{ (99\% reliability)}$$

$$\begin{aligned}
Z_n &= 2.466N^{-0.056} = 0.84 \\
C_H &= 1 + [(0.00889)(\frac{450}{375}) - 0.00829](\frac{61}{23} - 1) = 1.004 \\
S_{c, Grade 2} &= 2.41(375) + 237 = 1140.75 \\
\sigma_{c, allow} &= \frac{S_c Z_n C_H}{K_t K_r} = 962 \text{ MPa} \\
K_a &= 1 \quad K_s = 1 \quad K_m = 1.145843 \quad K_v = 1.45 \\
E_g &\approx 200 \text{ GPa} \quad \nu_g \approx 0.285 \quad E_p \approx 200 \text{ GPa} \quad \nu_p \approx 0.285 \\
K_E &= \sqrt{\frac{2}{\frac{1-\nu_p^2}{E_p} + \frac{1-\nu_g^2}{E_g}}} = 466563 \quad I = \frac{\pi \cos(20) \sin(20)}{1 + \frac{115}{305}} = 0.7332 \\
\sigma_c &= 466563 \sqrt{\frac{11174}{(0.06)(0.115)(0.7332)} (1)(1)(1.145843)(1.45)} = 857 \text{ MPa} \\
n_s &= \frac{962}{857} = 1.12
\end{aligned}$$

This gear design has a satisfactory contact stress factor of safety (greater than 1.1).

Bearings E and F Analysis

$$\begin{aligned}
n_4 &= 364.3 \text{ rpm} \quad m = 3 \text{ (for ball bearings)} \\
E_R &= 8258 \text{ N} \quad F_R = 3633 \text{ N} \quad E_R > F_R \Rightarrow P_r = 8258 \text{ N} \\
A_p &= 1 \quad X = 1 \quad P_a = 0 \quad P = P_r X = 8258 \text{ N} \\
L_{10} &= 318 \text{ million cycles} \\
C_{calc} &= 8258(318)^{\frac{1}{3}} = 56379 \text{ N}
\end{aligned}$$

From the bearing catalog: bearing SKF 6310, ID of 50 mm, OD of 110 mm, width of 27 mm and a specified load rating of 61800 N satisfies our needs for bearing E's radial load.

As bearing F has a lower radial load, for simplicity, ease of assembly, and maintenance, we will choose SKF 6310 for both bearings.